Software Test Plan

For: Online Quiz System

1. Introduction

This document defines the overarching software test plan for the Online Quiz System, adhering to the standard IEEE format. It clearly outlines the testing scope, testing approach, resource requirements, and test scenarios.

2. Test Items

- Frontend Web Application (React UI)
- Backend REST API Services (Node.js)
- Database Data Layer (PostgreSQL)

3. Features to be Tested

- User Authentication Workflows (FR1, SEC1)
- Role-based Access Control and Permissions (SEC2)
- Quiz Creation & Management by Teachers (FR2)
- Quiz Taking & Submission by Students (FR3)
- Automated Grading Calculation Accuracy (FR4)
- System Performance & Concurrency (NFR1, NFR2)

4. Features Not to be Tested

- Underlying hardware/network infrastructure and routing beyond the application's hosting environment.
- Third-party mailing systems for password resets (out of current phase scope).

5. Approach

Testing will include unit tests for core logical functions (like grading calculations), integration tests for database communication, and End-to-End (E2E) testing for UI-driven workflows. Performance testing will be conducted using automated load generation scripts.

5.1 Security Validation

Security testing will strictly focus on validating authentication flows, session management, and authorization rules. Specific tests will be run to attempt bypassing RBAC (Role-Based Access Control) boundaries. We will also execute tests for common vulnerabilities including SQL Injection (SQLi) and Cross-Site Scripting (XSS), and actively verify that the bcrypt password hashing mechanisms are correctly configured.

6. Traceability Aspects

Test Case ID	SRS Requirement ID	Requirement Description
TC-01, TC-02	FR1	User Authentication
TC-03	FR2	Quiz Creation
TC-04	FR3	Quiz Taking
TC-05	FR4	Automated Grading
TC-06	NFR1	Performance & Response Times
TC-07	SEC1	Cryptographic Password Hashing

Test Case ID	SRS Requirement ID	Requirement Description
TC-08, TC-09, TC-10	SEC2	Security & Access Control Policies

7. Test Cases

ID	Test Scenario	Preconditions	Steps	Expected Result	Pass/Fail
TC-01	Valid Student Login	Student account exists in DB	1. Navigate to login. 2. Enter valid email and password. 3. Click Login.	User is successfully authenticated and routed to Student Dashboard.	
TC-02	Invalid Login Attempt	Account exists in DB	1. Navigate to login. 2. Enter incorrect password. 3. Click Login.	System securely denies access and displays "Invalid credentials" error.	
TC-03	Teacher Creates Quiz	Logged in as Teacher	1. Click "New Quiz". 2. Enter Title and 1 Multiple-choice Question. 3. Click Save.	Quiz is saved to DB and immediately appears in the Teacher's quiz management list.	
TC-04	Student Submits Quiz	Logged in as Student	1. Open an active quiz. 2. Select answers for all questions. 3. Click Submit.	Submission is successfully recorded and the user is locked from retaking the quiz.	
TC-05	Auto-Grading Accuracy	Quiz submitted	1. Submit quiz with known pre-calculated correct	System automatically calculates and displays the exact 100% correct score based on the	

ID	Test Scenario	Preconditions	Steps	Expected Result	Pass/Fail
			answers. 2. Check the final result.	answer key.	
TC-06	Performance (Response)	System running under normal load	1. Submit a completed quiz. 2. Measure time until grade is shown.	The final graded result is displayed to the student in < 2.0 seconds.	
TC-07	Password Encryption	Active account exists	1. Inspect the Database 'users' table directly via DB client.	The password column displays a bcrypt hash, confirming no plain text passwords exist.	
TC-08	Enforce RBAC (Teacher)	Logged in as Student	1. Attempt to force-navigate to `/teacher/dashboard` via URL.	Access is denied (HTTP 403 Forbidden).	
TC-09	API Security (Auth)	Unauthenticated User	1. Send an anonymous POST request to `/api/quizzes`.	The API outright rejects the request returning HTTP 401 Unauthorized.	
TC-10	SQL Injection Prevention	On the public login page	1. Enter `' OR 1=1 --` into the email field. 2. Click Login.	Input is safely sanitized/parameterized; the system safely rejects the login attempt.	